



// IDP AS A SERVICE: MANAGING TENANTS EFFICIENTLY USING GITOPS

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Slides



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Version: 202511111539-3f1067e

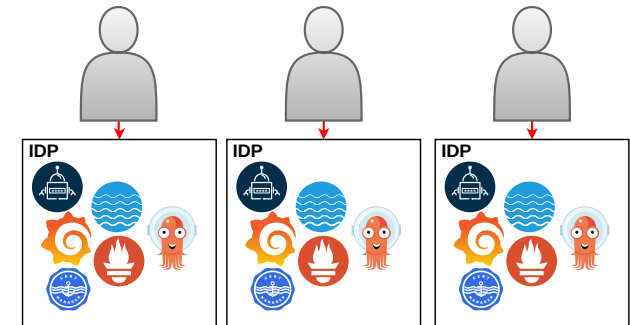
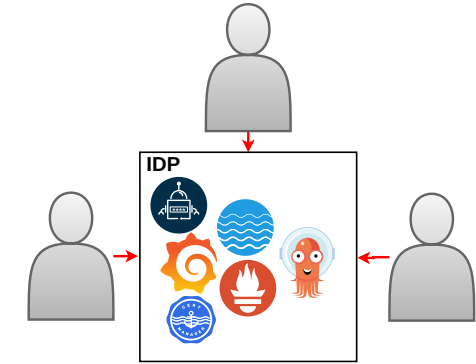
Agenda

- Shared IDP Instance
- Dedicated IDP Instances (IDPaaS)
- Multiple Environments per Tenant

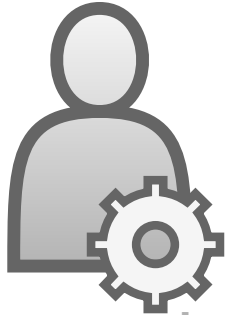


Multi-tenancy strategies

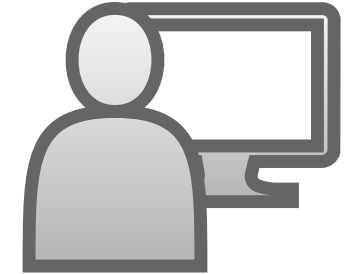
- Shared instance
- Dedicated instances



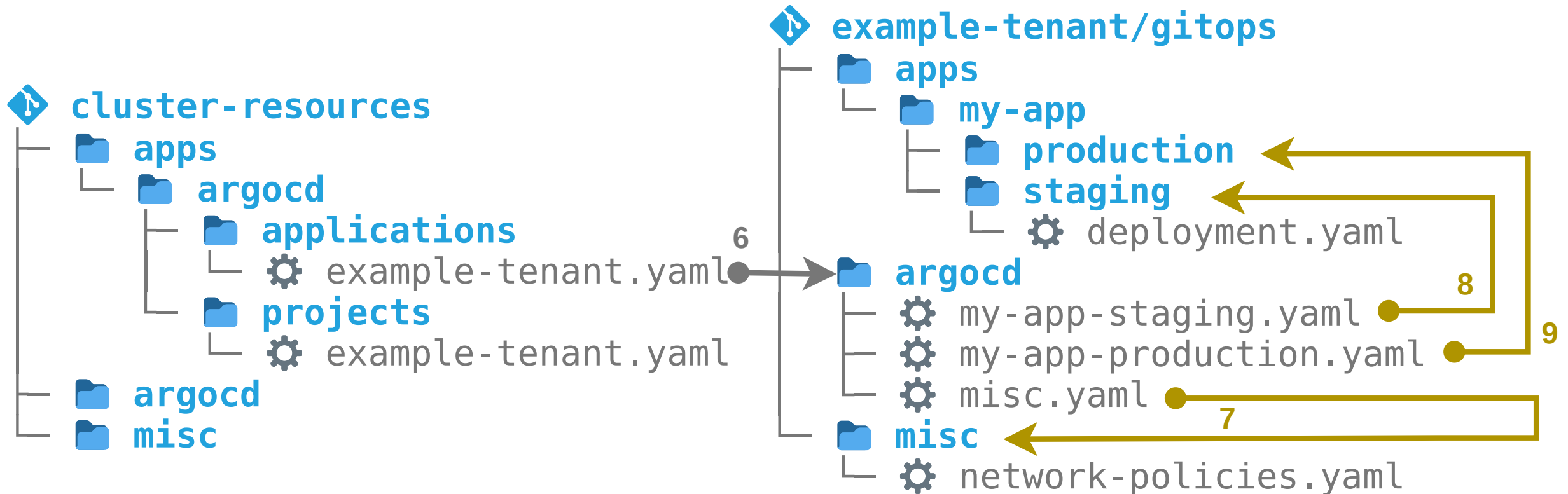
Shared Instance: A first approach



Platform engineer



Software engineer



Shared vs Dedicated Instance

- Easy to maintain
- Low infra usage
- Higher isolation
- Easier to configure 

 blog.argoproj.io/best-practices-for-multi-tenancy-in-argo-cd-273e25a047b0

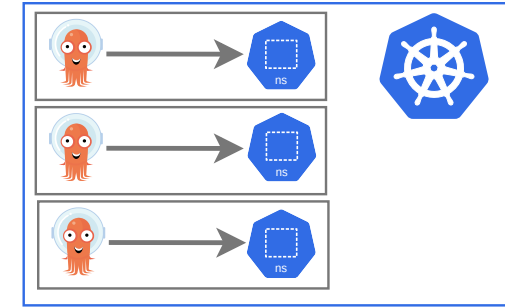
Dedicated instances: Options

Topology:

- Instance per Namespace
- Instance per Cluster

Management:

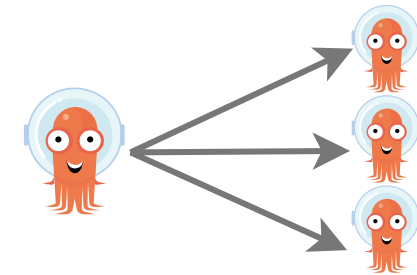
- Standalone
- Hub and Spoke (IDP as a Service)
 - 💡 Argo CD AppSets



Instance per Namespace

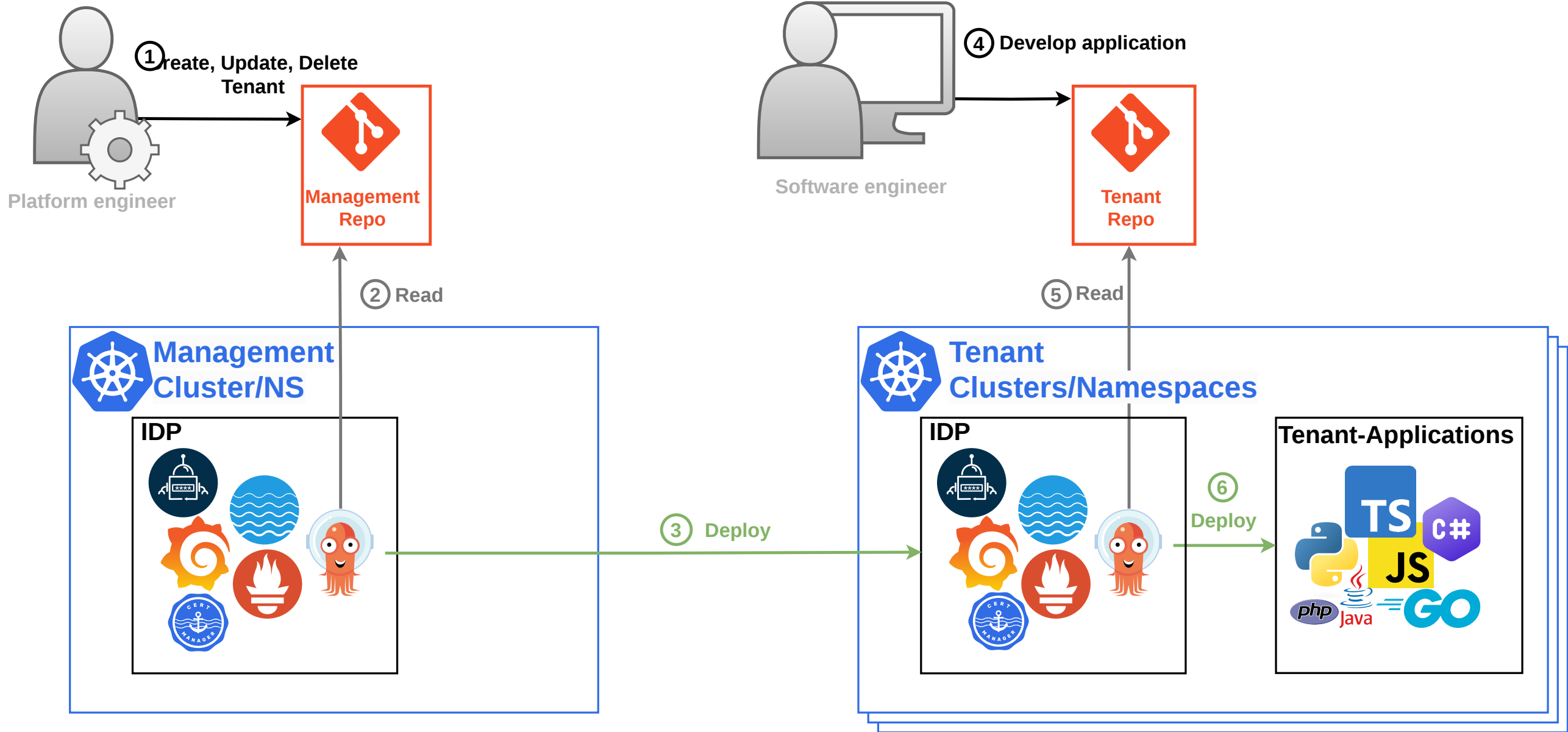


Instance per Cluster

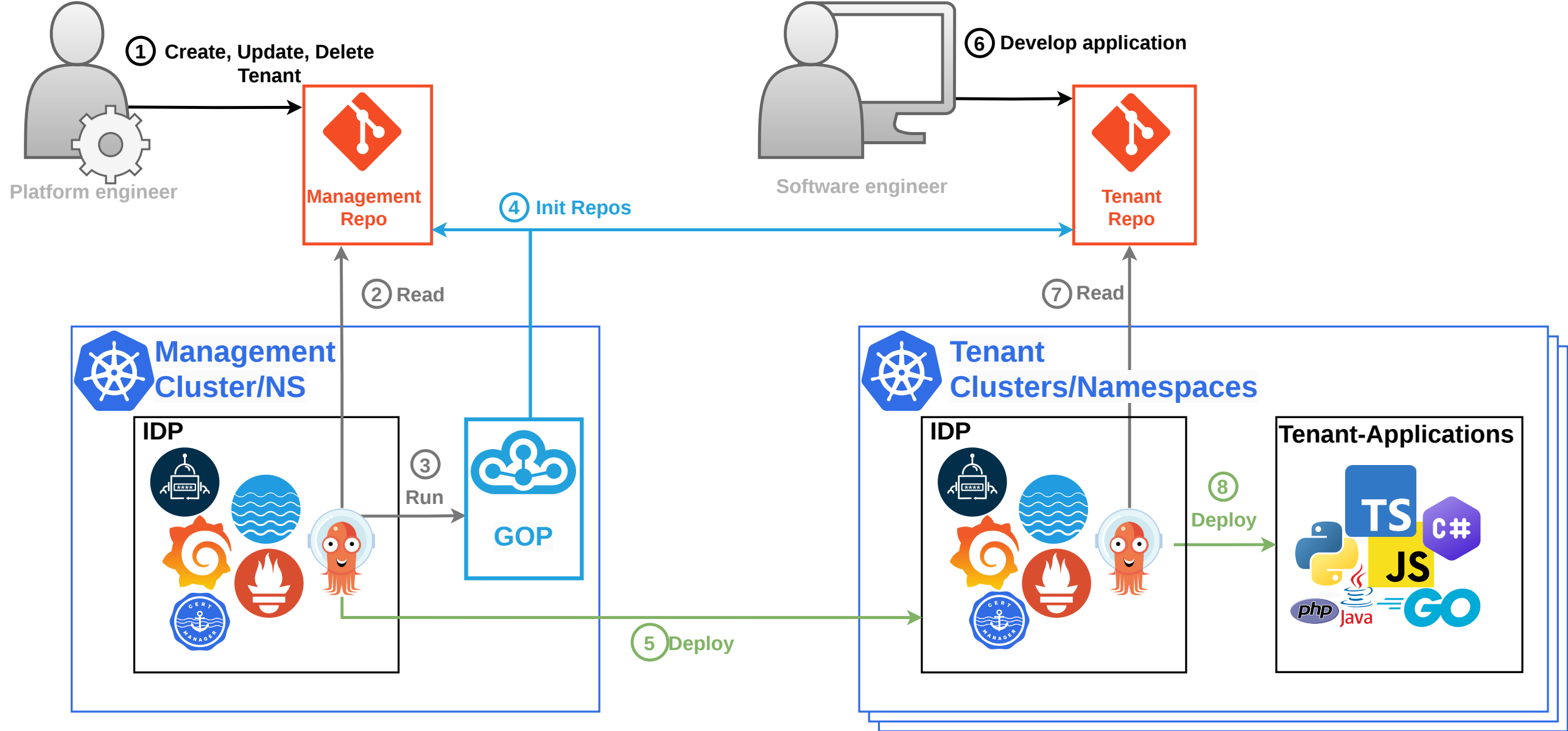


Hub and Spoke

IDP as a Service using Argo CD AppSets



IDP as a Service using Argo CD AppSets





1



2
Generate one app per file:
"/tenant-values.yaml"



tenants

tenant1

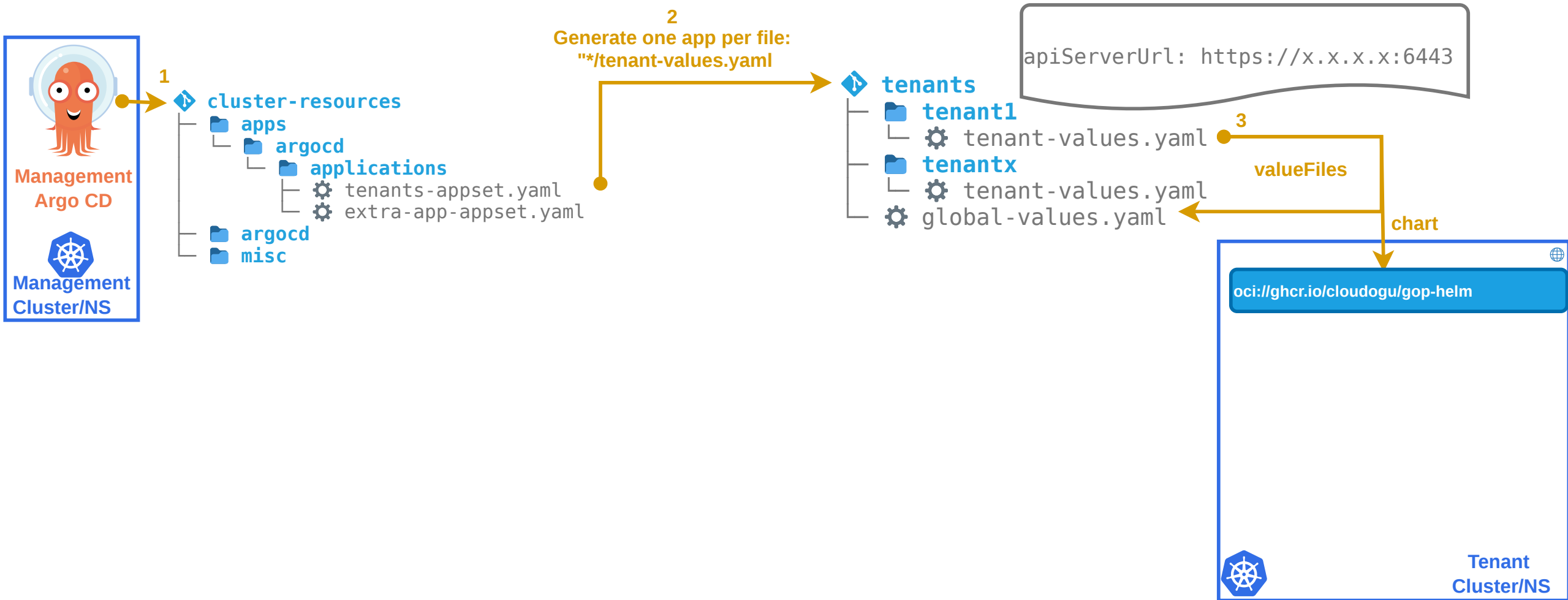
tenant-values.yaml

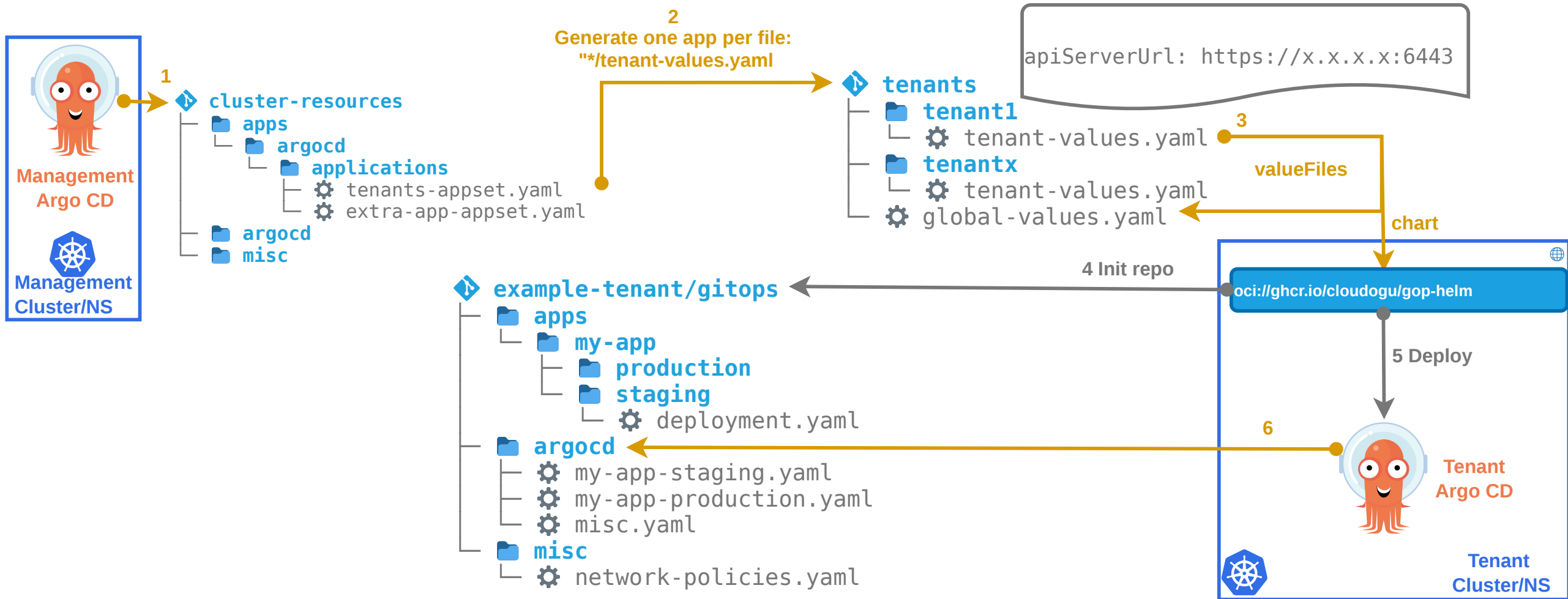
tenantx

tenant-values.yaml

global-values.yaml

apiServerUrl: https://x.x.x.x:6443



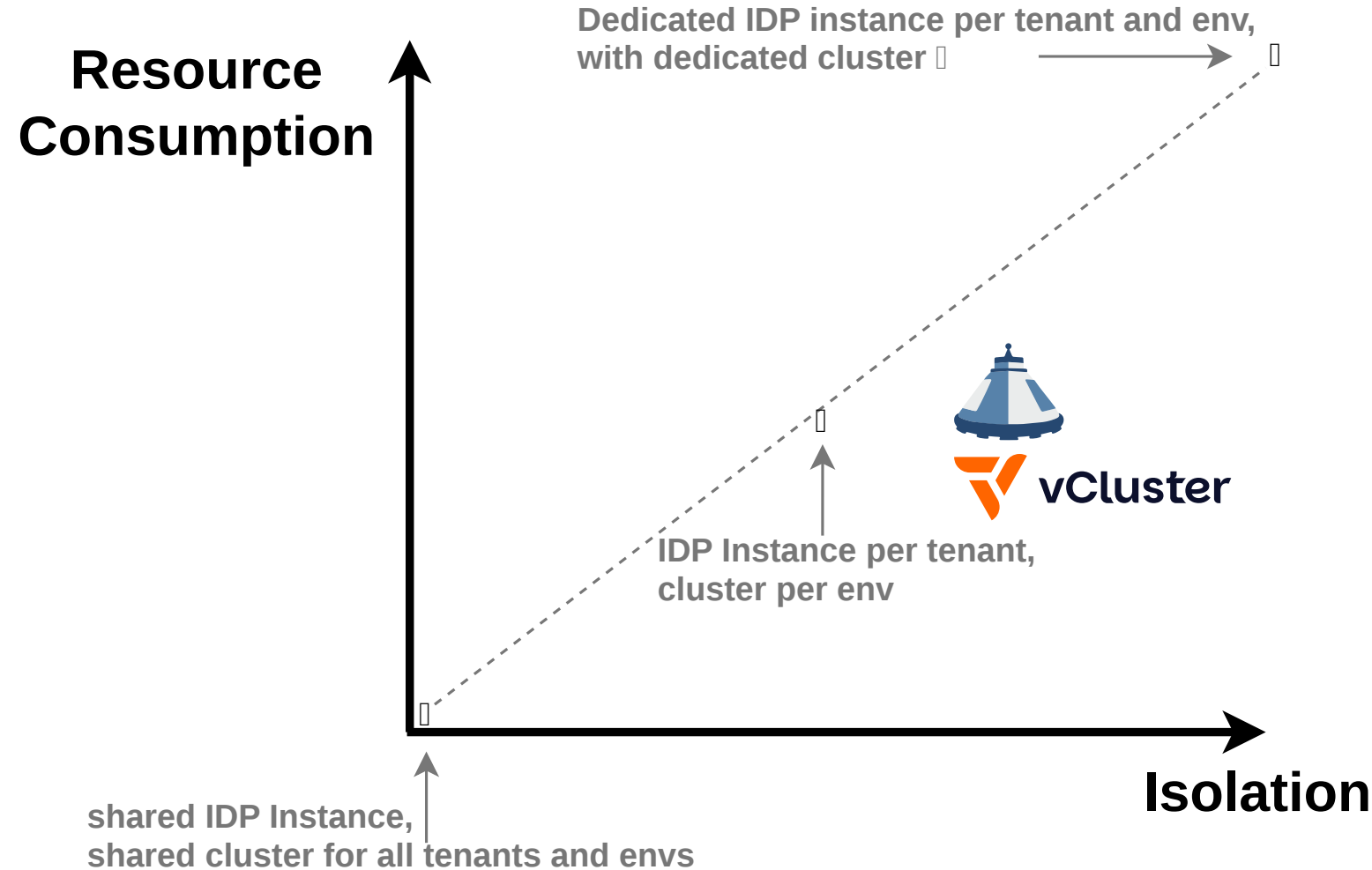


Instance per Namespace vs per Cluster

- Easy to maintain
- Low infra usage
- Higher isolation
- Easier to configure 

What about env/stages?

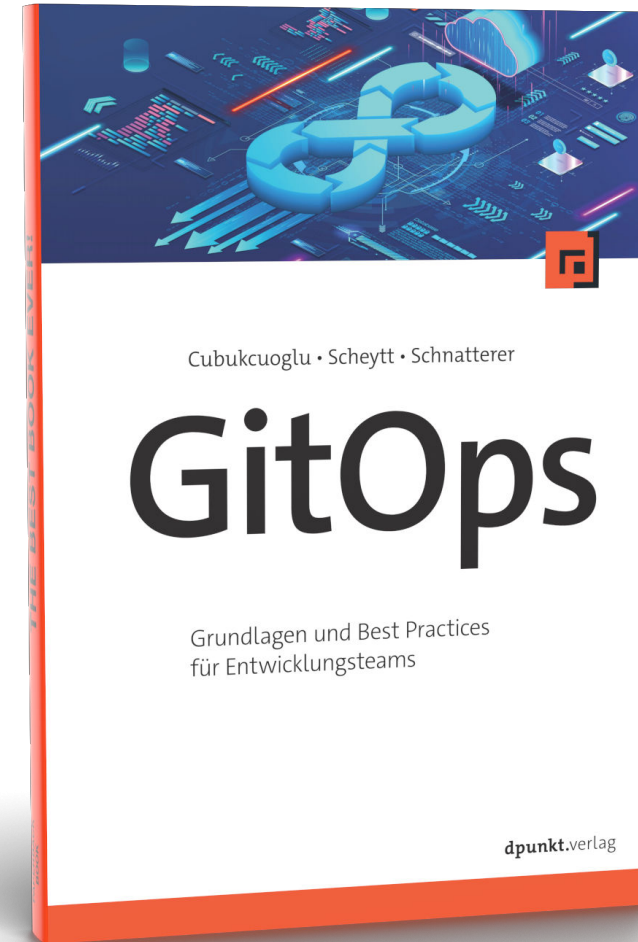
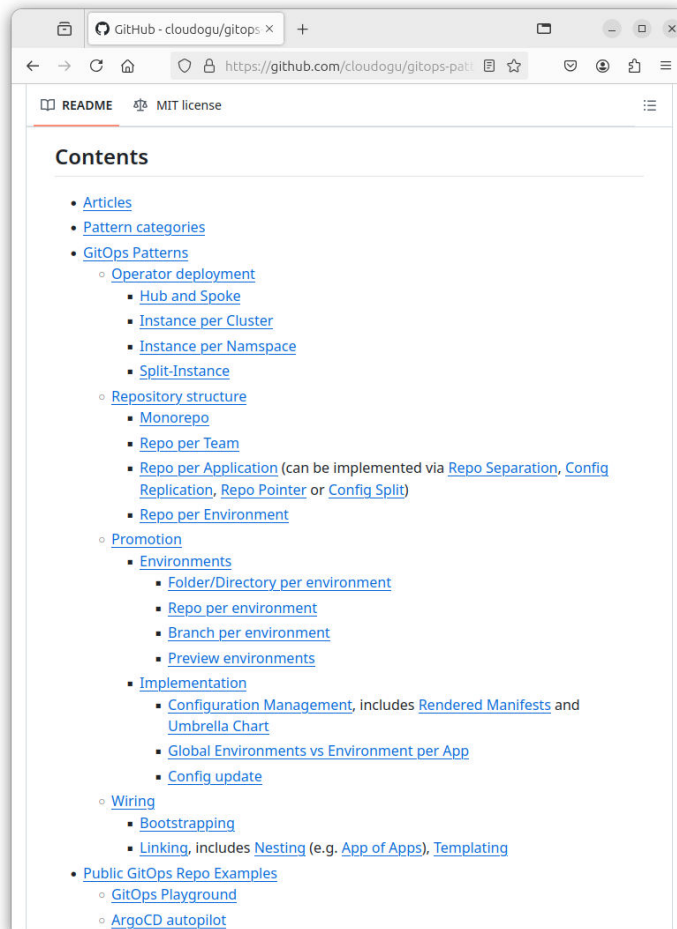
Each tenant might have multiple envs



Key Takeaways IDPs and tenants

- **Shared Instance:** Works in small settings
 - **Dedicated instances:** Higher isolation but more effort
- Minimize efforts implementing IDPaaS using AppSets 🐙
- Number of namespaces/clusters depends on requirements, e.g. envs

Further reading



 [cloudogu/gitops-patterns](https://github.com/cloudogu/gitops-patterns)

IDP Bootstrapping made simple: GOP

TODO Shared

Creates a complete GitOps-based operational stack / IDP on your Kubernetes clusters

 [cloudogu/gitops-playground](https://github.com/cloudogu/gitops-playground)

Also runs locally



```
VERSION='0.12.0'
bash <(curl -s \
  "https://raw.githubusercontent.com/cloudogu/gitops-playground/$VERSION/scripts/init-cluster.sh") \
  && docker run --rm -t -u $(id -u) \
    -v ~/.config/k3d/kubeconfig-gitops-playground.yaml:/home/.kube/config \
    --net=host \
    ghcr.io/cloudogu/gitops-playground:$VERSION --yes --argocd --ingress-nginx --base-url=http://localhost
# More IDP-features: --monitoring --vault=dev --cert-manager --mailhog
# More features for developers: --jenkins --registry --content-examples
```

Try with GOP:  [cloudogu/gop-multi-tenant-multi-cluster-example](https://github.com/cloudogu/gop-multi-tenant-multi-cluster-example)

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Slides



☒ Let's meet



 clouddogu/gitops-playground

Image sources

- Live Example:
<https://unsplash.com/photos/assorted-color-hot-air-balloons-during-daytime-DuBNA1QMpPA>
- More examples:
<https://unsplash.com/photos/XZc4f2XZc84>
- coloured-parchment-paper background by brgfx on Freepik:
https://www.freepik.com/free-vector/coloured-parchment-paper-designs_1078492.htm